

Treatments applied for biosecurity purposes are part of measures used in managing the risk of introduction of exotic pests and diseases. These treatments are only effective when conducted correctly.

It is important that treatment providers in Australia (onshore) and other countries (offshore) understand and follow our treatment requirements so that Australia remains free of exotic pests and diseases. Consignments that do not comply with treatment requirements may not be allowed entry into Australia.

Our methodologies outlining best practice for applying biosecurity treatments, and document templates for recording and certifying compliance, are available below.

Applications for approval of new phytosanitary treatments should follow the guidance provided below.

Methodology

The heat treatment methodology sets out the minimum requirements for treatment providers performing heat treatments on commodities and/or associated packaging for quarantine and pre-shipment purposes.

Documentation

BMSB treatment providers should use the below templates for BMSB heat treatment certification and records.

Summary of key changes to the heat treatment methodology

The heat treatment methodology was updated in March 2021. The following table details the key changes:

Section	Changes
Scope	Removed reference to “core” temperature
General	Replaced terminated with withdrawn
1.4.2	The entire target of heat treatment must be enclosed under the sheet
Certificate example	Remove BMSB title and added container as target option

Methodology and Guide

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If you have difficulty accessing these files, [contact us](#) for help.

BMSB heat treatment certificate and record of treatment templates

Download

- [BMSB Heat treatment certificate – template \(DOCX 93 KB\)](#)
- [How to complete a BMSB heat treatment certificate \(PDF 25 KB\)](#)
- [How to complete a BMSB heat treatment certificate \(DOCX 32 KB\)](#)
- [BMSB Record of heat treatment – template \(DOCX 144 KB\)](#)

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Generic heat treatment certificate and record of treatment templates

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Methodology

The Methyl bromide fumigation methodology sets out the minimum requirements for treatment providers performing methyl bromide fumigations on commodities and/or associated packaging for quarantine and pre-shipment (QPS) purposes.

You can find a list of the requirements in section 12.

Documentation

Methodology

Download

- [Methyl bromide fumigation methodology v3.0 \(PDF 1.7 MB\)](#)
- [Methyl bromide fumigation methodology v3.0 \(DOCX 3.7 MB\)](#)

If you have difficulty accessing these files, [contact us](#) for help.

Methyl bromide treatment certificates and treatment record templates

We have provided these templates and certificates as a resource to help you prepare your own treatment documentation. Use of these templates is optional.

Download

- [Fumigation certificate \(generic\) methyl bromide v3.0 – template \(DOCX 66 KB\)](#)
- [Fumigation certificate \(BMSB\) methyl bromide v3.0 – template \(DOCX 48 KB\)](#)
- [Record of fumigation \(non-perishables\) methyl bromide v3.0 – template \(DOCX 64 KB\)](#)
- [Record of fumigation \(perishables\) methyl bromide v3.0 – template \(DOCX 64 KB\)](#)

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Helpful documentation

Download

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Guide to performing fumigation with methyl bromide

This guide provides information on the various methods and techniques that can be used to ensure that fumigators perform fumigations that meet the requirements of the methodology.

This document should be read in conjunction with the Methyl bromide fumigation methodology.

Download

- [Guide to performing QPS fumigations with methyl bromide v3.0 \(PDF 3.1 MB\)](#)
- [Guide to performing QPS fumigations with methyl bromide v3.0 \(DOCX 5.8 MB\)](#)

If you have difficulty accessing these files, [contact us](#) for help.

Guide to packaging suitability for performing QPS treatments

This guide provides information on meeting consignment suitability requirements for fumigation and controlled atmosphere treatments.

Download

- [Guide to packaging suitability for performing QPS treatments v3.0 \(PDF 1.5 MB\)](#)
- [Guide to packaging suitability for performing QPS treatments v3.0 \(DOCX 141 MB\)](#)

If you have difficulty accessing these files, [contact us](#) for help.

Fact sheets

These fact sheets guide you through the documentation requirements for methyl bromide fumigation.

Download

If you have difficulty accessing these files, [contact us](#) for help.

The Controlled atmosphere treatment methodology sets out the minimum requirements for treatment providers performing controlled atmosphere treatments on commodities and/or associated packaging for quarantine and pre-shipment (QPS) purposes.

To ensure the controlled atmosphere treatment methodology is practical and fit for purpose, a draft was released for public consultation on the department's website between 23 August and 12 September 2021. All feedback received was considered in the development of the final version of the methodology.

Summary of changes to the controlled atmosphere treatment methodology from the consultation draft:

- clarifications to language and amendments to typographical errors
- changes to pressure testing requirements for sheeted enclosures
- changes to requirements for the use of a vaporiser when applying carbon dioxide and nitrogen gas to an enclosure
- addition of minimum accuracy requirements for gas concentration monitoring equipment
- addition of a definition for pressure testing to the glossary.

This document should be read in conjunction with the Guide to packaging suitability for performing QPS treatments. While not intended to specifically cover controlled atmosphere treatments, some of the concepts discussed in the Guide to performing QPS fumigations with methyl bromide are transferable. The guide can be found under the [methyl bromide fumigations section of this page](#).

Methodology

Download

- [Controlled atmosphere treatment methodology \(PDF 960 KB\)](#)
- [Controlled atmosphere treatment methodology \(DOCX 1.6 MB\)](#)

If you have difficulty accessing these files, [contact us](#) for help.

Controlled atmosphere treatment certificates and treatment record templates

Download

- [Controlled atmosphere treatment certificate – template \(PDF 404 KB\)](#)
- [Record of controlled atmosphere treatment – template \(DOCX 69 KB\)](#)

If you have difficulty accessing these files, [contact us](#) for help.

Methodology

The sulfuryl fluoride fumigation methodology sets out the minimum requirements for BMSB treatment providers performing sulfuryl fluoride fumigations on commodities and/or associated packaging for quarantine and pre-shipment purposes.

Documentation

BMSB treatment providers should use the below templates for their sulfuryl fluoride fumigation certification and records.

Note: there are separate certificate and treatment record templates for treatment providers operating under a distributors stewardship program* using a third party approved program and for those that are not using an approved third party approved program.

*The approved stewardship program operators and their third party programs are:

- Douglas Products or their agents – Fumiguide
- Ensystex II, Inc (USA) – Fumicalc
- Barcan Pest Control & Fumigation Co. (Barcan ilaclama Fumigasyon Tic. Ltd.Sti.) (Turkey) - FumiTrack

Methodology

Download

- [Sulfuryl fluoride fumigation methodology \(PDF 1.1 MB\)](#)
- [Sulfuryl fluoride fumigation methodology \(DOCX 776 KB\)](#)

If you have difficulty accessing these files, [contact us](#) for help.

BMSB sulfuryl fluoride treatment certificate and treatment records templates

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If you have difficulty accessing these files, [contact us](#) for help.

Methodology

The Ethyl formate fumigation methodology sets out the minimum requirements for treatment providers performing ethyl formate fumigations on commodities and/or associated packaging for quarantine and pre-shipment (QPS) purposes.

Documentation

Methodology

Download

If you have difficulty accessing these files, [contact us](#) for help.

Generic ethyl formate treatment certificates and treatment record templates

Download

- [Ethyl formate treatment certificate \(PDF 128 KB\)](#)
- [Ethyl formate treatment certificate \(DOCX 44 KB\)](#)
- [Ethyl formate record of fumigation \(PDF 163 KB\)](#)
- [Ethyl formate record of fumigation \(DOCX 64 KB\)](#)

If you have difficulty accessing these files, [contact us](#) for help.

Methodology

The Insecticide treatment methodology sets out the minimum requirements for treatment providers performing insecticide treatments on commodities, containers and/or associated packaging suited to such treatments for Quarantine and Pre-shipment (QPS) purposes. This methodology is the basis for compliance auditing of treatment providers to monitor their performance of effective treatments with insecticides.

Documentation

Insecticide treatment providers should use the below templates for their insecticide treatment certification and records.

Methodology

Download

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Guide

Download

- [Guide to treating sea containers with deltamethrin \(PDF 571 KB\)](#)
- [Guide to treating sea containers with deltamethrin \(DOCX 1.2 MB\)](#)
- [Insecticide Treatment Certificate Guide \(PDF 548 KB\)](#)
- [Insecticide Treatment Certificate Guide \(DOCX 510 KB\)](#)

If you have difficulty accessing these files, [contact us](#) for help.

Insecticide treatment certificate and treatment and calibration records templates

Download

- [Record of Insecticide treatment – template \(DOCX 35 KB\)](#)
- [Record of Calibration – template \(DOCX 15 KB\)](#)

- [Insecticide treatment certificate – template \(DOCX 33 KB\)](#)

If you have difficulty accessing these files, [contact us](#) for help.

Insecticide Application Calculator

Specified treatment rate

Treatment applied

Treatment applied

g/L to % converter

Calculated requirements

If you are having difficulty accessing this tool, you may download the [offline version](#).

Treatment provider instructions

- Confirm the specified treatment rate.
- Enter the number of containers being treated next to the corresponding container size.
Please note- this calculator will only work for standard 20” and 40” containers.
For all other containers, please refer to offshoretreatments@aff.gov.au.
- Enter the Concentration of the active constituent. This can be found on your product label.

The calculated requirements will now be populated in the yellow box. This information should be used to prepare for the treatment.

Further information on insecticide treatments certificate requirements can be found on the Insecticide Treatment Certificate Guide.

Treatment certificate assessment instructions

- Enter the number of containers being treated next to the corresponding container size.
Please note- this calculator will only work for standard 20” and 40” containers.
For all other containers, please refer to offshoretreatments@aff.gov.au.
- Enter the Concentration of the active constituent. This can be found in the “Insecticide Details” field. If concentrate is expressed in g/L, then please use to g/L to % calculator provided. This will trigger the concentration to calculate.
- Enter all remaining fields from the “Treatment applied” section of the certificate.
- Check that all fields are compliant.

For non-compliant certificates, please refer to offshoretreatments@aff.gov.au.

For all broker related enquiries, please contact aepsupport@aff.gov.au.

Further information on insecticide treatments certificate requirements can be found on the Insecticide Treatment Certificate Guide.

Download

If you have difficulty accessing these files, [contact us](#) for help.

Applications for approval of new phytosanitary treatments should follow the following guidance. Applications can be sent to offshoretreatments@aff.gov.au.

For applications for new treatments for seeds for sowing, please see the specific information on the [imported seeds for sowing](#) page.

International standards

Data submissions should aim to address the requirements included in the International Standards for Phytosanitary Measures ([ISPM 28](#)) Phytosanitary treatments for regulated pests.

Information on the pests and commodity targets of the proposed treatment

The application should clearly indicate whether the proposed phytosanitary treatment is for a single pest or for multiple pests. This requirement also applies to proposed treatments for pests not covered by ISPM requirements, for example, contaminating pests, hitchhiking pests, or pests that are not plant pests.

The application should clearly indicate whether the proposed phytosanitary treatment is for a single commodity or for multiple commodities.

Based on the above 2 points, the application should identify whether the proposed treatment is for:

- a single pest on a single commodity
- a single pest on multiple commodities
- multiple pests on a single commodity
- multiple pests on multiple commodities.

The application should provide accepted taxonomic identification of the target pests, to species level, or in the case of multiple pests, all species to the taxon level, for example, all species of coleoptera, or all species of arthropods.

Information on the proposed treatment

The application should include information on:

- the biological mode of action of the proposed phytosanitary treatment on the target taxa
- the physical behaviour of the treatment in the target commodities
- the application method, i.e. how the proposed phytosanitary treatment should be applied

- the monitoring method, i.e. how the correct application of the treatment will be monitored and recorded, for example, placement of electronic gas sensors and/or temperature sensors in goods
- information on the final product, i.e. the form of its actual use
 - the formulation or the mix of chemicals
 - dosage
 - duration of exposure
 - temperature during treatment
 - Ct (Concentration x time).
- information on the experimental methodology used in studies conducted for developing the proposed phytosanitary treatment. Note: Generally, the same standard of literature review, experimental design, data recording, data analysis and confidence levels sufficient for a peer reviewed publication would be expected in applications, including provision of the data used for the analyses. This information should include
 - how test organisms were exposed during experimentation (e.g., application method and the testing environment such as containers used)
 - experimental design
 - number of experimental units (e.g., number of insects)
 - number of replicates and information on control treatments
 - whether experiments consisted of laboratory trials and/or field trials.
- data on dose-response mortality, including mortality curves
- numbers of experimental units for testing should target efficacy standards accepted by NPPOs for phytosanitary measures, which are usually very high assurance levels of mortality. Assurance requirements will vary based on the likely consequence of the pests and the size of infestations that may be typically encountered in the target commodities. However, phytosanitary treatment requirements are always going to need higher assurance levels than treatments applied to 'control' pests.
- information on the most resistant life stage/s of the target pest/s that are considered likely to be in the target commodities
- information on other effects or side effects of the treatment, such as likely effects of the treatment on commodities, residues in food, etc.
- Work, Health and Safety information related to the phytosanitary application of the treatment (this is not required for initial efficacy assessment but is needed prior to the final approval by the department).
- Information on the application methodology and training for applicators for the proposed treatment (not required for the initial efficacy assessment but is needed if the treatment is to be applied onshore in Australia).
- published and unpublished references and data.

Disclaimers:

This information outlines data requirements for the efficacy assessment of new phytosanitary treatments. It is not intended to provide information on the department's or treatment applicator's

operational readiness to apply new treatments, nor any other matters unrelated to the efficacy assessment that are required for other approval processes.

All chemical treatments must be also approved by the Australian Pesticides and Veterinary Medicines Authority (APVMA) before they can be applied to commodities in Australia.

All chemical treatments must also have residue approval from Food Standards Australia New Zealand (FSANZ) before they can be used on food commodities to be sold in Australia.

We may change these conditions if circumstances require, such as in emergency situations or changes related to global issues.

Source: <https://www.agriculture.gov.au/biosecurity-trade/import/arrival/treatments/treatments-fumigants>